

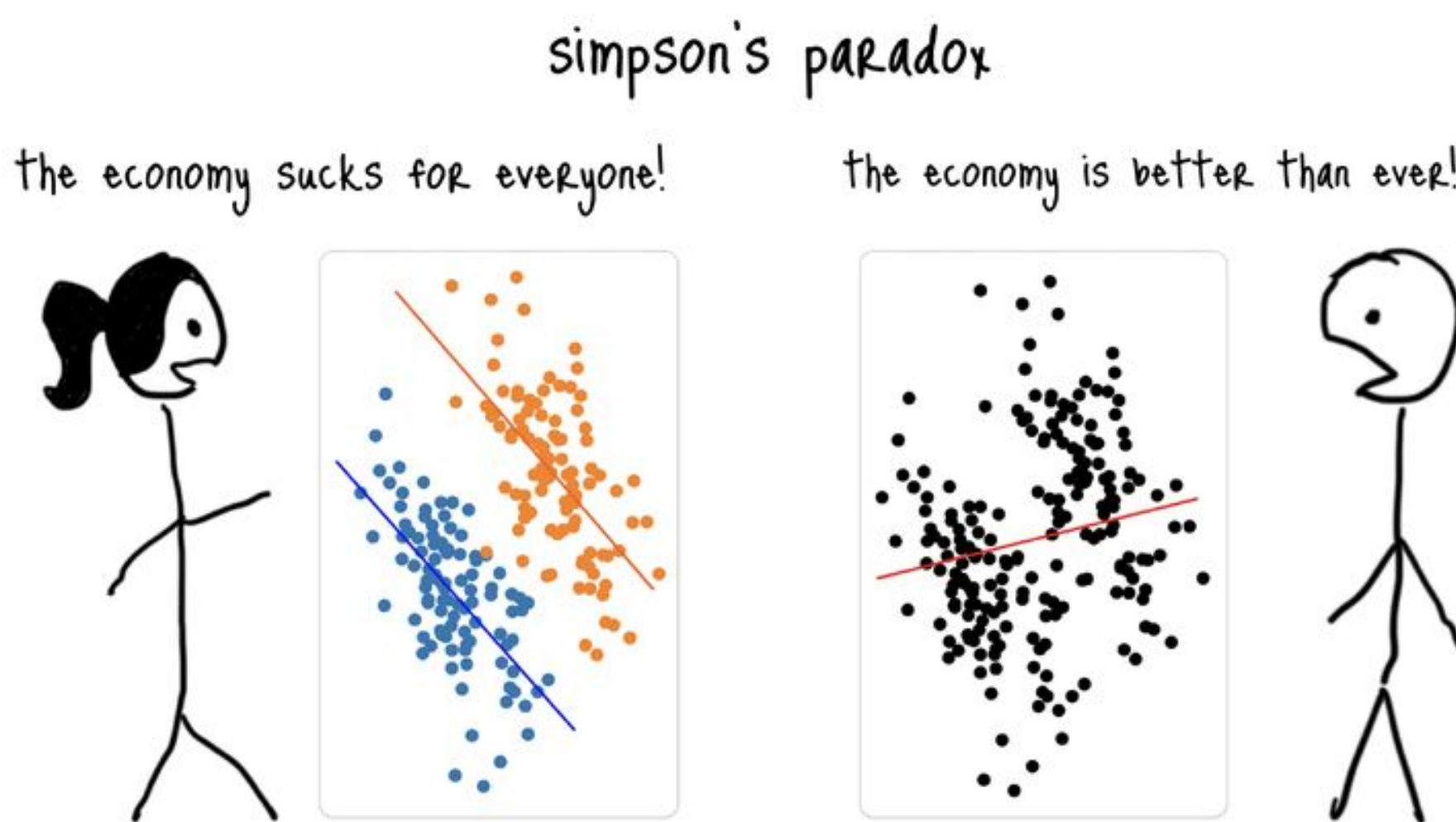


The Problem

You fit a regression on the full dataset. R^2 is fine, p-values are small. A reviewer asks: “Does this hold for each subgroup?”

You check. It doesn’t. Now what?

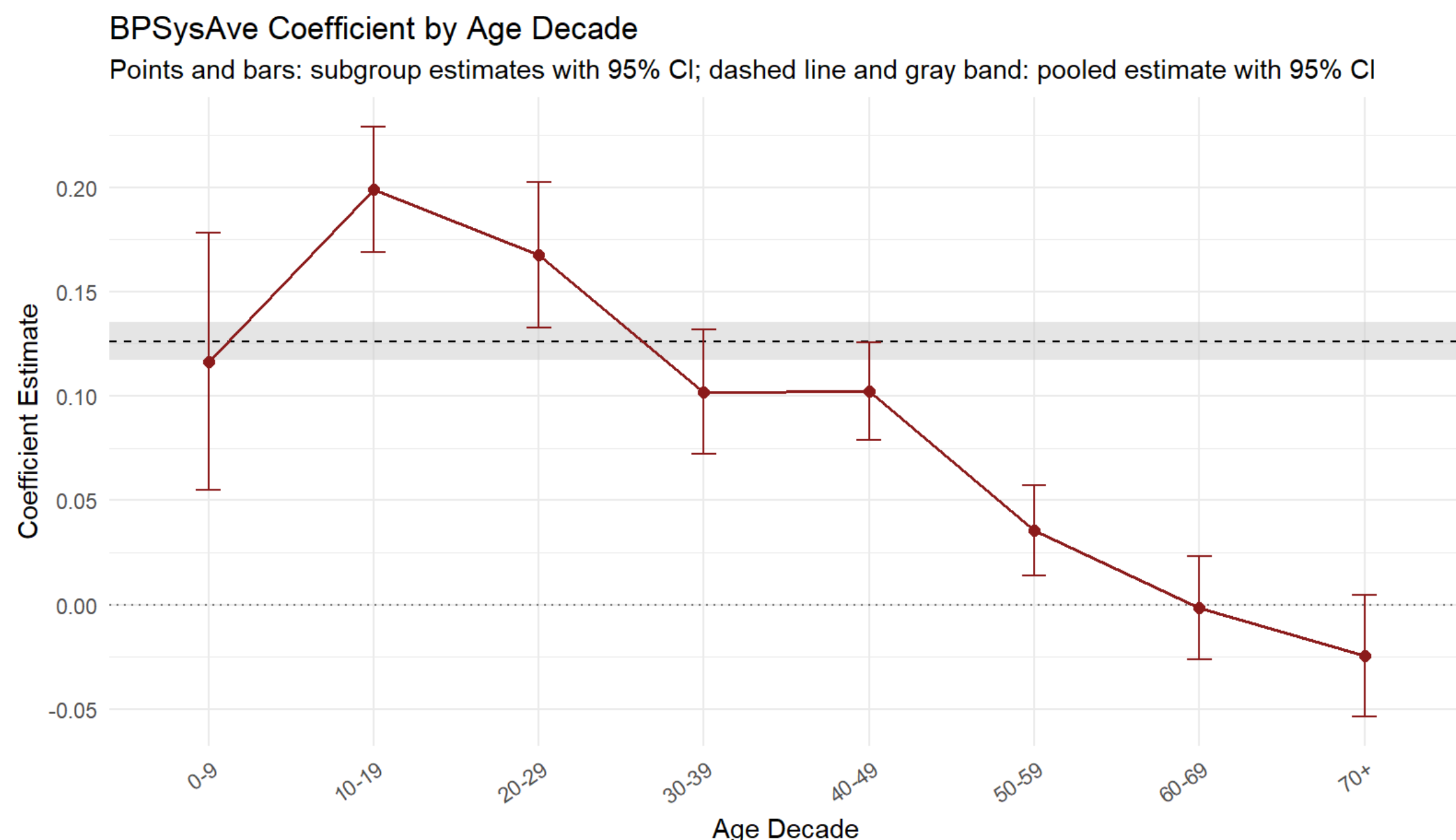
Simpson’s Paradox



Within each group the trend is positive. Pooled, it reverses. The pooled regression hides subgroup structure.

Real-Data: NHANES BPSysAve

NHANES 2017–2018 ($n = 9,254$). Response: systolic BP. Subgroups: age decades. Focus predictor: BPSysAve.



Metric	Value
SSR (BPSysAve)	0.57
Std. drift	32.07
Context-norm. drift	7.49

Key finding: Effect *reverses sign* in older groups. Drift 32→7.5 after context normalization, confirming genuine heterogeneity.

Four Models (nested)

M1: pooled $Y = \beta_0 + X^\top \beta + \varepsilon$

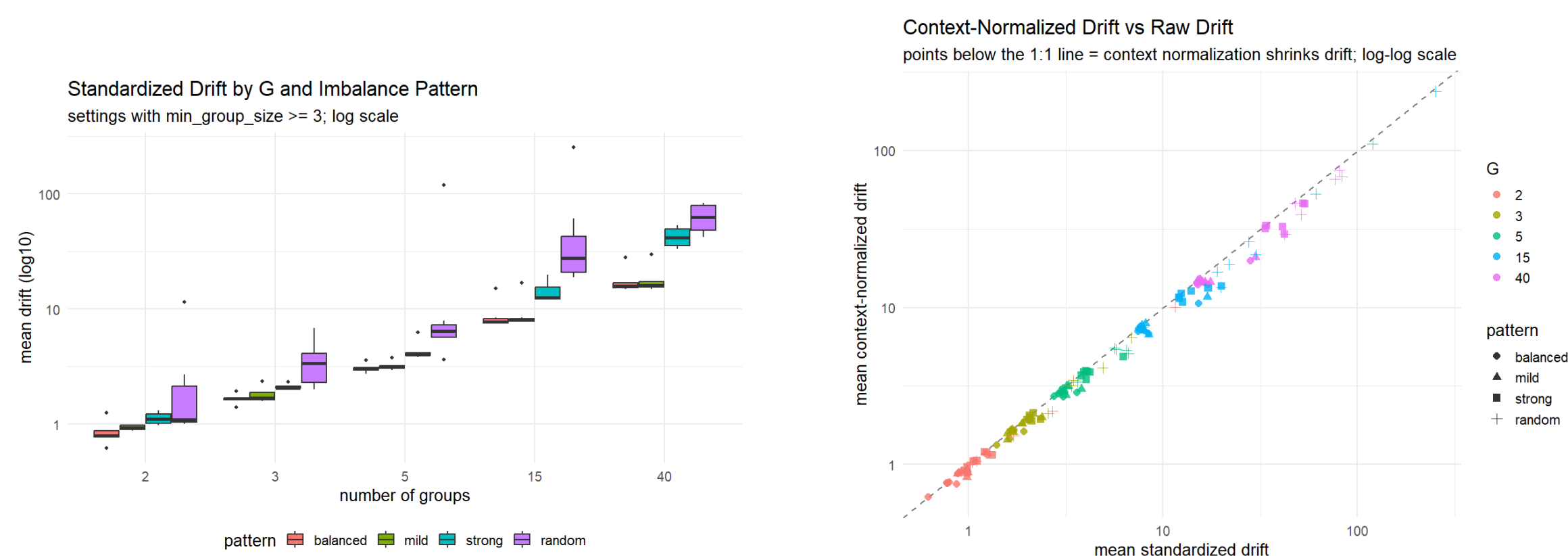
M2: add group intercept γ_G

M3: add interactions $X^\top \delta_G$

M4: fully separate model per group

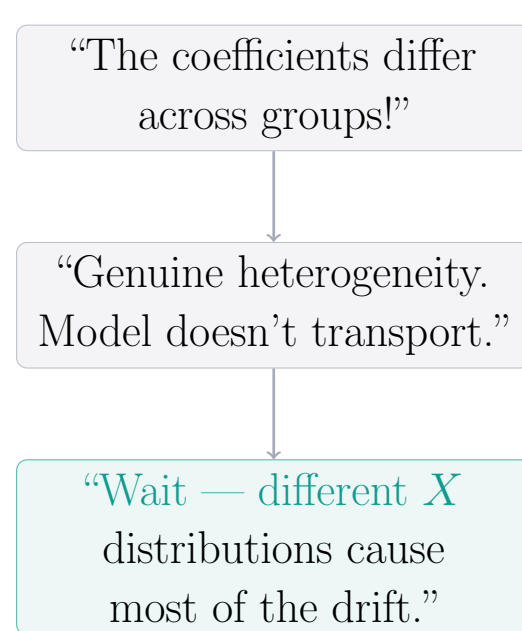
Simulation Grid (Scenario 1)

140 settings: $G \in \{3, 5, 10, 20, 40\}$, $n \in \{100, \dots, 30000\}$, 4 imbalance patterns (balanced, mild, strong, Dirichlet-random). 200 reps each.



Drift grows with G and imbalance (left). Context normalization shrinks drift (right).

“It’s an Artifact”



Coefficient differences come from **two sources**: genuine heterogeneity and mechanical artifacts from covariate distribution differences. Standard practice conflates them.

Five Stability Metrics

Metric	Definition	What it captures
SSR	$SSR_j = \frac{1}{M} \sum \mathbf{1}(\text{sgn } \hat{\beta}_{j,a} = \text{sgn } \hat{\beta}_{j,b})$	Do conclusions (+/–) reverse across groups?
Standardized drift	$D_j^* = \frac{\max_g \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} }{\text{SE}(\hat{\beta}_{j,\text{pooled}})}$	How far do subgroup coefficients move from pooled, in SE units?
Context-norm. drift	$\tilde{D}_j = \frac{D_j^*}{1 + M_j}$, $M_j = \frac{1}{\binom{G}{2}} \sum_{a < b} \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	Drift after accounting for covariate distribution differences.
TPG	$TPG_{a \rightarrow b} = \text{RMSE}_{\text{pooled} \rightarrow b} - \text{RMSE}_{b \rightarrow b}$	Prediction loss when a model travels across groups.
CSS	$CSS_{a,b} = \frac{1}{p} \sum_j \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	How different are X distributions between groups?

Key Decomposition: Why Context Drift < Raw Drift

Even with identical β , different X distributions inflate drift:

$$\hat{\beta}(g) - \hat{\beta}(\text{pooled}) = (X_g' X_g)^{-1} X_g' \varepsilon_g - (X' X)^{-1} X' \varepsilon$$

$$D_j^* = \underbrace{\tilde{D}_j}_{\text{Total}} \times (1 + M_j)$$

3 regimes: $M_j \approx 0 \Rightarrow$ all drift real; large M_j + small $\tilde{D}_j \Rightarrow$ artifact; both large $\Rightarrow$ genuine heterogeneity.

What Drives Instability?

- Drift driven by G and per-group n , not total N .
- Strong/random imbalance amplifies drift 2–4× over balanced.
- SSR = 1.0 throughout (Scenario 1: identical β , no sign flips).
- Context normalization reduces drift; shrinkage $\propto$ CSS.

Future Work

Same analysis for Scenarios 2–4, then:

- Introduce **collinearity** across predictors
- Scale to **higher dimensions** (more predictors)
- Test **different regression models** beyond OLS

References

Berk et al. (2019); Tsvi et al. (2026); Bertsimas & Paskov (2020); Liu et al. (2020); Watson & Engle (1985); Garbade (1977); Buja et al. (2006).
github.com/fadjoukiiii/Regression-conclusion-stability-paper